

AIM

To write a C program to perform basic string operations such as copying one string to another, comparing two strings, reversing a string, and removing spaces from a string without using the string.h header file.

PROCEDURE

1. Start the program.
2. Declare character arrays to store input string, copied string, reversed string, and space-removed string.
3. Read the input string from the user.
4. Copy the characters of the input string into another string using a loop.
5. Compare the two strings character by character to check whether they are equal.
6. Calculate the length of the string and reverse it by storing characters in reverse order.
7. Traverse the string and copy only non-space characters to remove spaces.
8. Display the results of all string operations.
9. End the program.

CODE

```
/*  
Filename : string.c  
Author   : Monish Kanna K  
Purpose  : To perform string operations such as copying,  
           comparing, reversing a string and removing spaces.  
*/  
  
#include <stdio.h>  
  
int main()  
{  
    char str1[100], str2[100];  
    char rev[100], nospace[100];  
    int i, j, len = 0;  
    int flag = 0;  
    printf("Enter a string: ");  
    scanf("%[^\\n]", str1);
```

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i = 0;
while (str1[i] != '\0')
{
    str2[i] = str1[i];
    i++;
}
str2[i] = '\0';
printf("\nCopied string: %s", str2);
i = 0;
while (str1[i] != '\0' && str2[i] != '\0')
{
    if (str1[i] != str2[i])
    {
        flag = 1;
        break;
    }
    i++;
}
if (flag == 0)
    printf("\nStrings are equal");
else
    printf("\nStrings are not equal");
i = 0;
while (str1[i] != '\0')
{
    len++;
    i++;
}
for (i = len - 1, j = 0; i >= 0; i--, j++)
{
    rev[j] = str1[i];
}
rev[j] = '\0';

```

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printf("\nReversed string: %s", rev);
i = 0;
j = 0;
while (str1[i] != '\0')
{
    if (str1[i] != ' ')
    {
        nospace[j] = str1[i];
        j++;
    }
    i++;
}
nospace[j] = '\0';
printf("\nString without spaces: %s", nospace);
return 0;
}

```

OUTPUT

```

PROBLEMS  TERMINAL  OUTPUT  PORTS
● PS C:\AMI C PROGRAM> cd "c:\AMI C PROGRAM\Monish_kanna_week2_practice1\" ; if ($?) { gcc string.c -o string } ; if ($?) { .\string }
Enter a string: monish kanna

Copied string: monish kanna
Strings are equal
Reversed string: annak hsinom
String without spaces: monishkanna

```

RESULT

The program correctly performed string copy, string comparison, string reversal, and removal of spaces without using the string.h library.